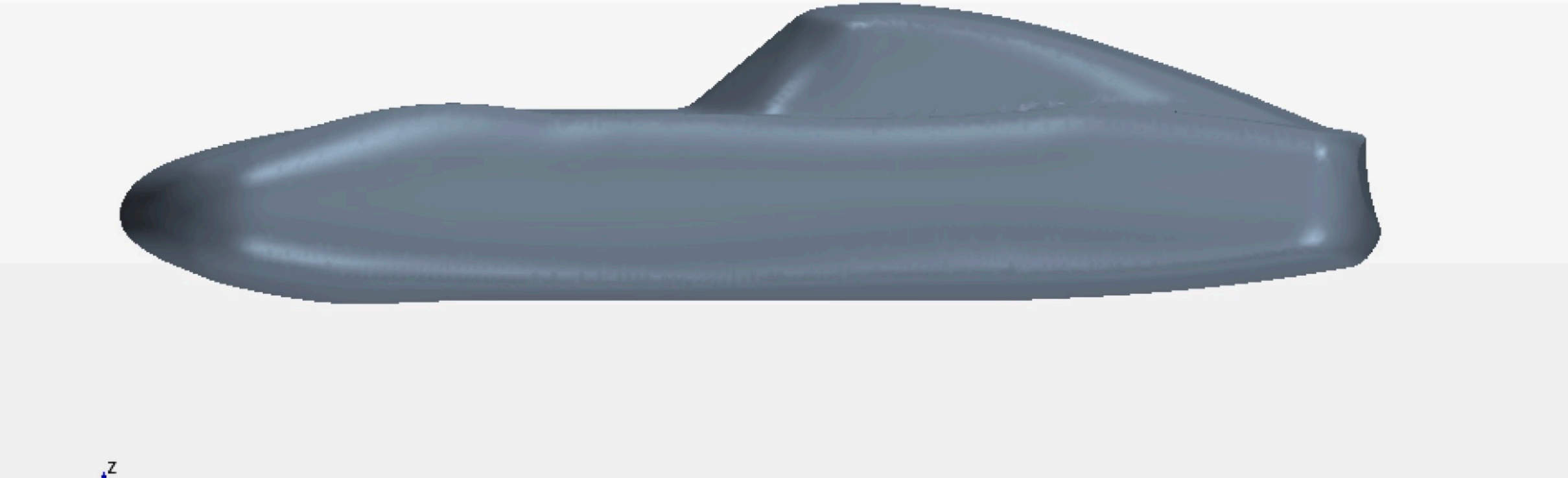
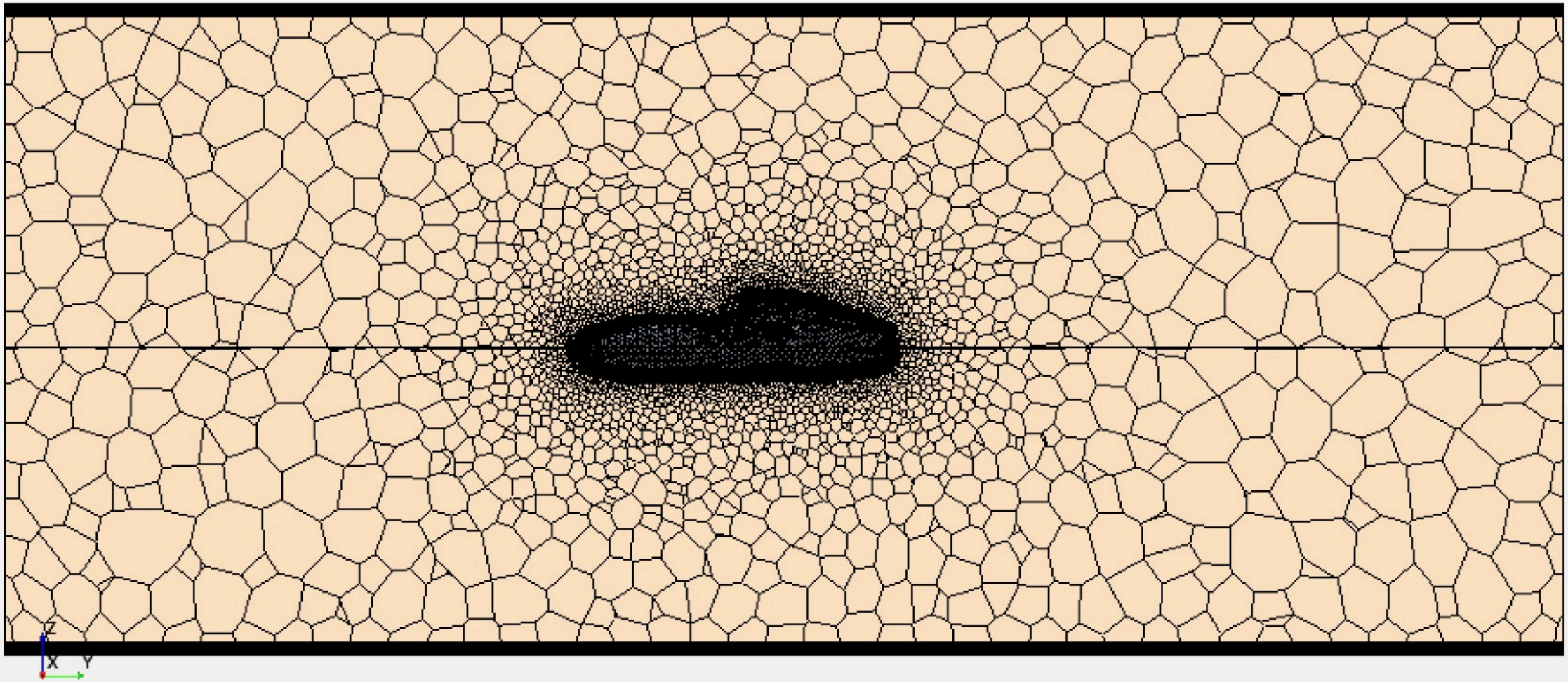
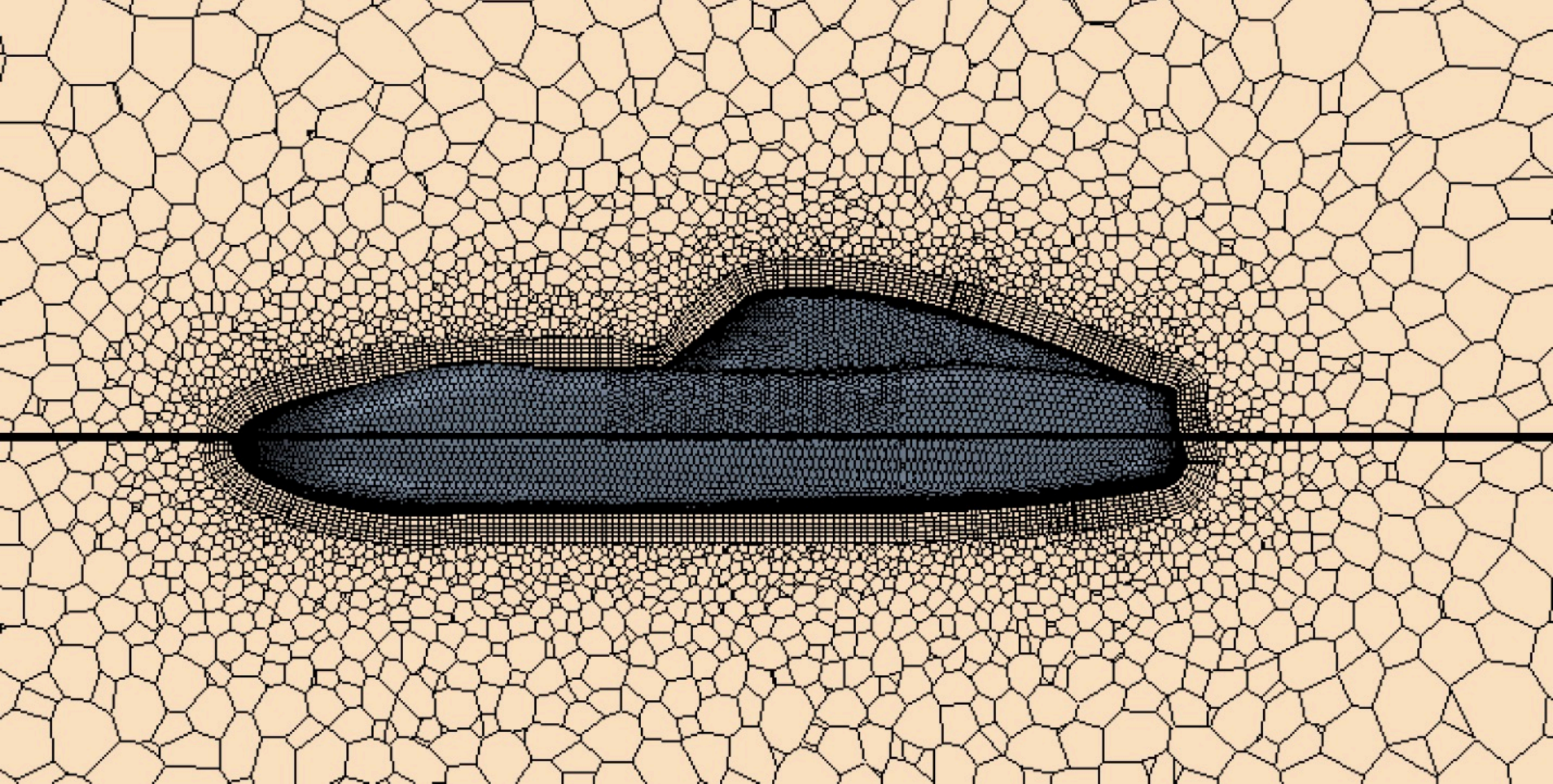
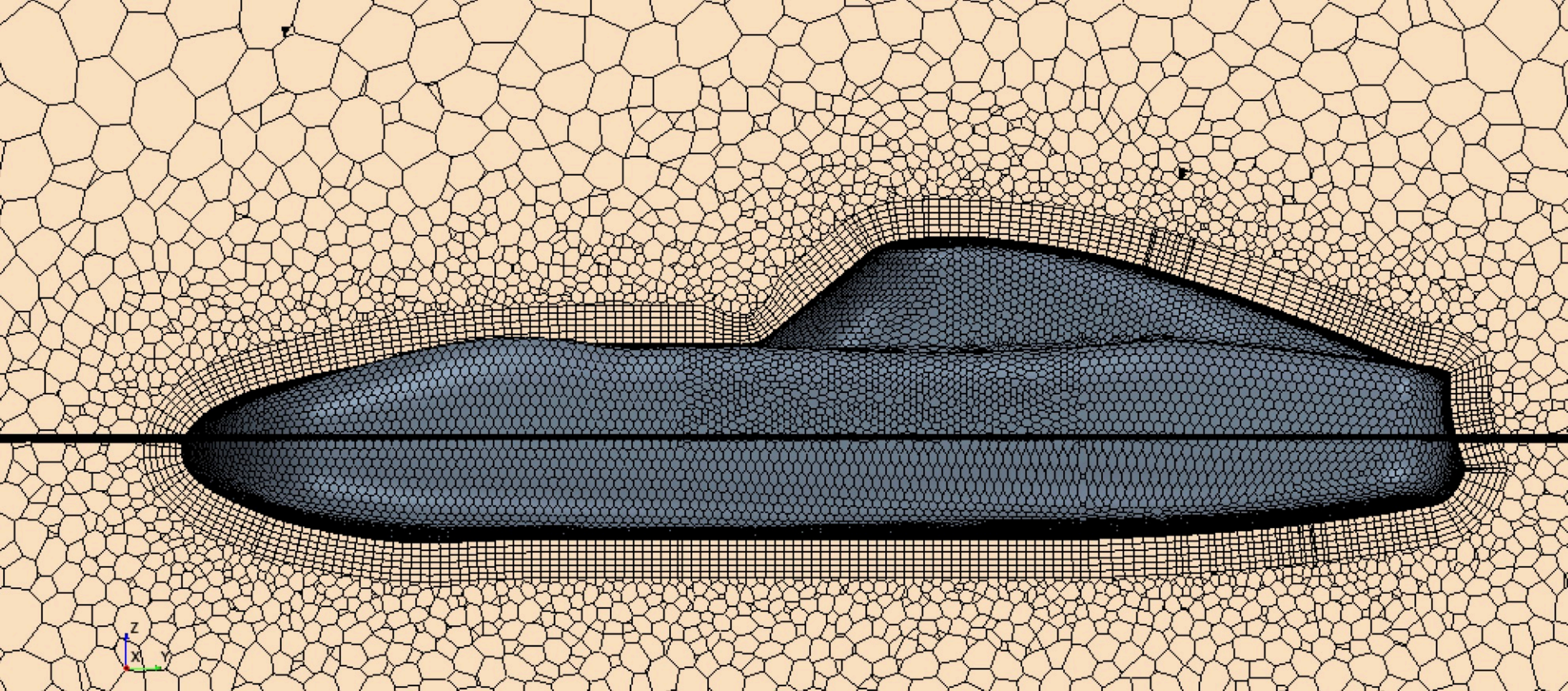
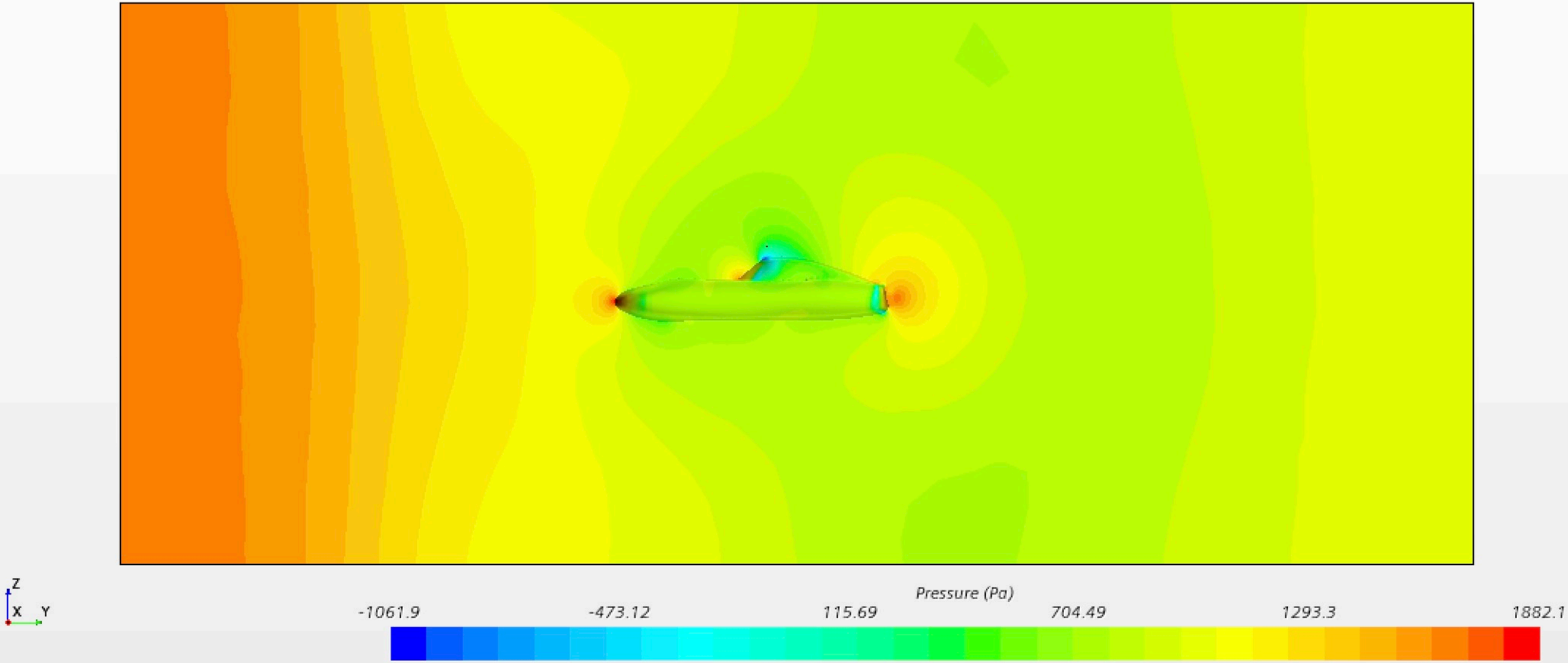
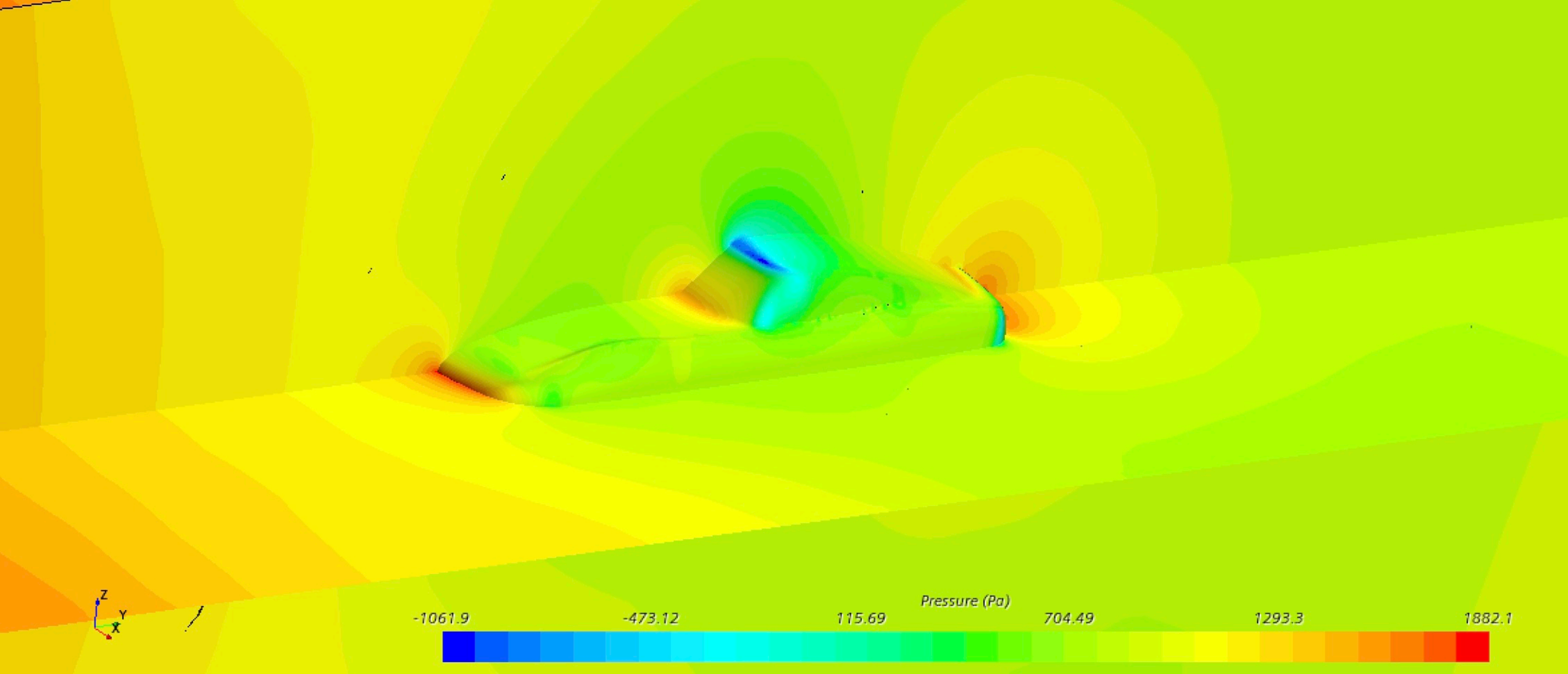

Z

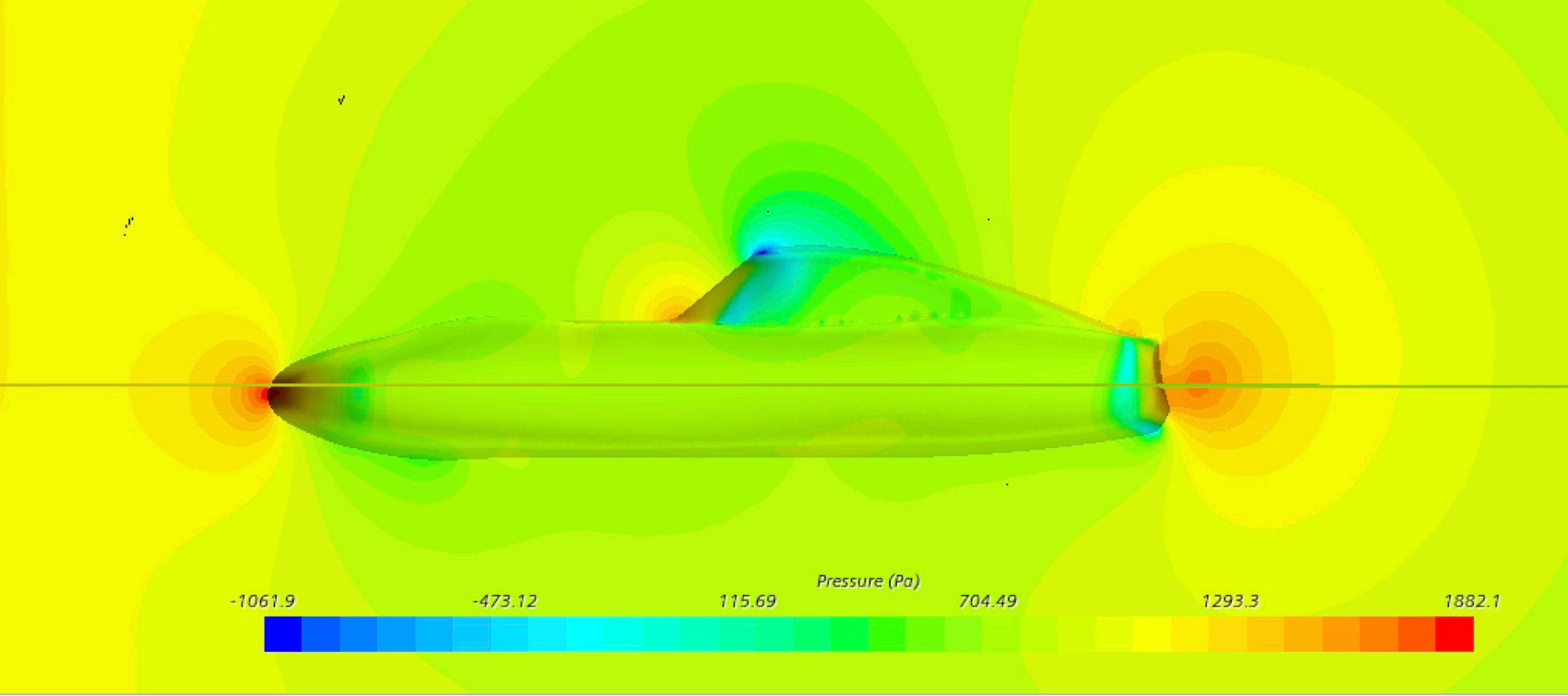

Z
X
Y

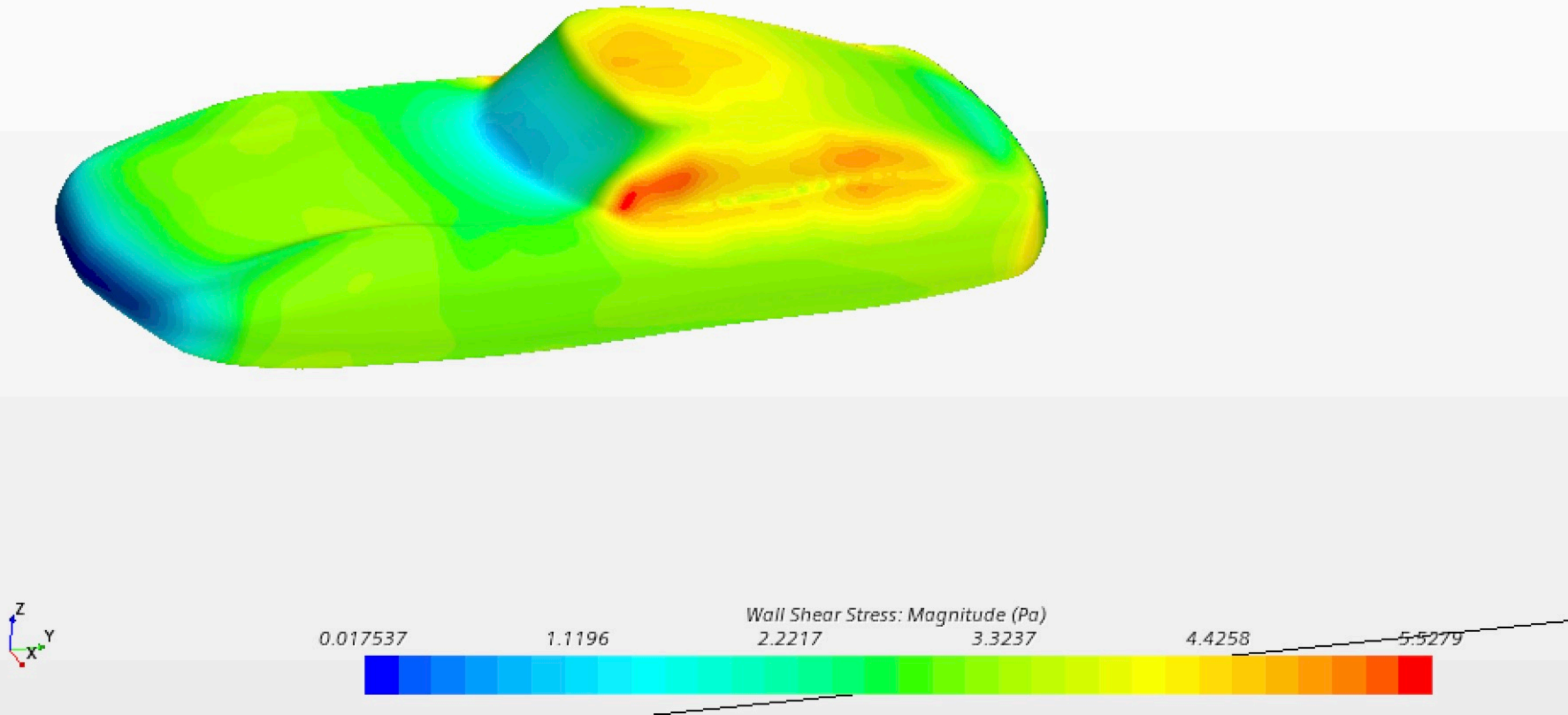


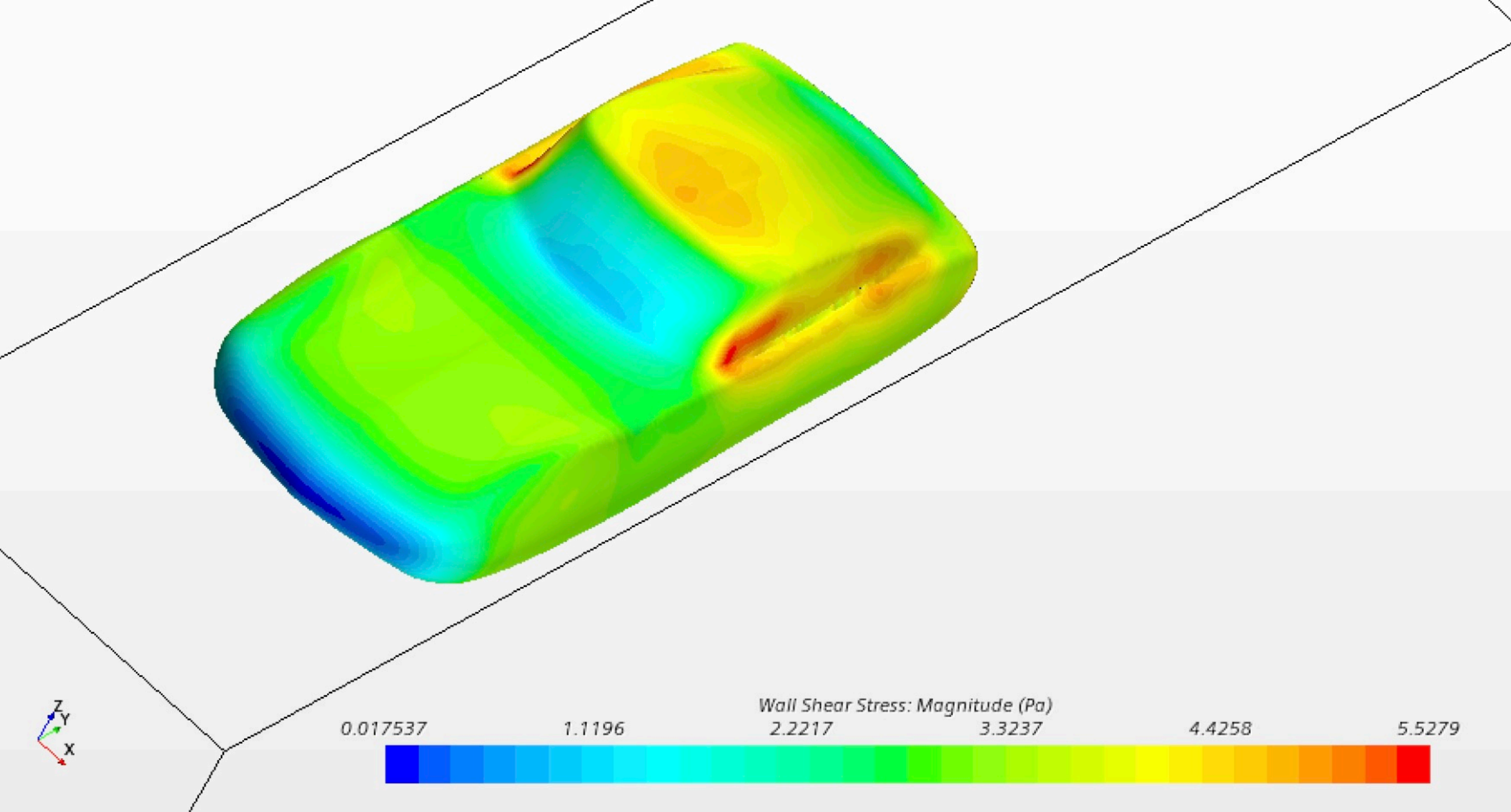

Z
X
Y


Z
X
Y
Pressure (Pa)
-1061.9
-473.12
115.69
704.49
1293.3
1882.1


Pressure (Pa)
-1061.9
-473.12
115.69
704.49
1293.3
1882.1
Z
Y
X


Pressure (Pa)
-1061.9
-473.12
115.69
704.49
1293.3
1882.1


Z
Y
X
Wall Shear Stress: Magnitude (Pa)
0.017537
1.1196
2.2217
3.3237
4.4258
5.5279


Wall Shear Stress: Magnitude (Pa)
0.017537
1.1196
2.2217
3.3237
4.4258
5.5279
Z
Y
X

Summary Report: Toyota Gt

Session Summary

Date	Feb 22, 2026, 11:49:20 AM
Simulation	Z:\Star CCM +\Toyota Gt.sim
File size	1.7e+02 MB
Number of Partitions	2
Number of Restored Partitions	2

Software Summary

Version	BuildArch: win64 BuildEnv: intel18.3vc14-r8 PresentationVersion: 2020.2.1 ReleaseDate: Thu Jul 16 15:33:18 UTC 2020 ReleaseNumber: 15.04.010
MPI Version	IBM Platform MPI-9.1.4.4

Hardware Summary

Hosts	Number Processes: 2 Rank[0]: VAMSHIKRISH8130 Rank[1]: VAMSHIKRISH8130

Simulation Properties

Toyota Gt		
Continua	Continua	2
Parts Meshes	Regions	[Car]
	Interfaces	[]
	Tags	[]
Physics 1	Regions	[Car]
	Interfaces	[]
	Point Sets	[]
	Active	true
	Tags	[]
Models		
Gas		
Air	Database Material	Air (Air) [Standard/Gases]
	Tags	[]
Material Properties		
Dynamic Viscosity	Method	Constant
Constant	Value	1.85508E-5 Pa-s
Molecular Weight	Method	Constant
Constant	Value	28.9664 kg/kmol
Specific Heat	Method	Constant
Constant	Value	1003.62 J/kg-K
Thermal Conductivity	Method	Constant
Constant	Value	0.0260305 W/m-K
Turbulent Prandtl Number	Method	Constant
Constant	Value	0.9
Gradients	Limiter Method	Venkatakrishnan
	Custom Accuracy Level Selector	2.0
	Verbose	false
	Least-Squares Quality Criterion	true

		Flat Cells Curvature Criterion	true
		Cell Skewness Criterion	true
		Chevron-Cell Criterion	true
		Least-Squares Tensor Minimum Eigenvalues Ratio	0.1
		Normalized Flat Cells Curvature Factor	1.0
		Maximum Safe (Positive) Skewness Angle (deg)	75.0
		Minimum Unsafe (Positive) Skewness Angle (deg)	88.0
		Use TVB Gradient Limiting	false
		Acceptable Field Variation (Factor)	0.05
	Gravity		
	Ideal Gas	Incompressible	false
		Density Limiting	false
	Implicit Unsteady	Continuum Iteration	37
		Continuum Physical Time	0.013000000000000005
	K-Epsilon Turbulence		
	Realizable K-Epsilon	Buoyancy Production of Dissipation	Boundary Layer Orientation
Two-Layer		Cmu	0.09
		C1e	1.44
		C2e	1.9
		Ct	1.0
		Sigma_k	1.0
		Sigma_e	1.2
		Sarkar	2.0
		Tke Minimum	1.0E-10
		Tdr Minimum	1.0E-10
		Secondary Gradients	On
		Convection	2nd-order
		Normal Stress Term	false
		Curvature Correction Option	Off
		Two-Layer Type	Shear Driven (Wolfstein)
		Two-Layer ReY*	60.0
		Two-Layer Delta ReY	10.0
	Reynolds-Averaged		
Navier-Stokes			
	Segregated Flow	Positivity Rate Limit	0.2
		Flow Boundary Diffusion	true
		Unsteady Flux Dissipation Corrections	false
		Limit Acoustic-CFL Option	Per-Model
		Secondary Gradients	On
		Convection	2nd-order
		Delta-V Dissipation	Off
	Segregated Fluid	Continuum Temperature	300.0 K
Isothermal			
	Solution Interpolation	Per Part Mapping	false
		Interpolation Method	Nearest neighbor
		Legacy Method	false
		Conservation Correction	Disable
	Three Dimensional Turbulent		
	Two-Layer All y+ Wall	Iterative Ustar	false
Treatment			
	Wall Distance	Wall Distance Method	Implicit Tree
Reference Values			
	Gravity	Value	[0.0, 0.0, -9.81] m/s^2

Reference Altitude	Value	[0.0, 0.0, 0.0] m
Reference Density	Value	1.0 kg/m^3
Minimum Allowable Wall Distance	Value	1.0E-6 m
Maximum Allowable Absolute Pressure	Value	1.0E8 Pa
Minimum Allowable Temperature	Value	100.0 K
Minimum Allowable Absolute Pressure	Value	1000.0 Pa
Maximum Allowable Temperature	Value	5000.0 K
Reference Pressure	Value	101325.0 Pa
Initial Conditions		
Pressure	Method	Constant
Constant	Value	900.0 Pa
Turbulence Intensity	Method	Constant
Constant	Value	0.5
Turbulence Specification	Method	Intensity + Viscosity Ratio
Turbulent Velocity Scale	Method	Constant
Constant	Value	1.0 m/s
Turbulent Viscosity Ratio	Method	Constant
Constant	Value	9.0
Velocity	Method	Constant
	Coordinate System	Laboratory
Constant	Value	[0.0, 40.0, 0.0] m/s
Regions	Part Selection Priority	[Car]
	Regions	1
Car	Index	0
	Mesh Continuum	Parts Meshes
	Physics Continuum	Physics 1
	Parts	[Subtract]
	Type	Fluid Region
	Allow Per-Part Values	false
	Tags	[]
Boundaries	Part Surface Selection Priority	[Car: Subtract.TOYOTA GT.Faces, Car: Subtract.Block.Top&Bottom, Car: Subtract.Block.Inlet, Car: Subtract.Block.outlet, Car: Subtract.Block.Sides]
	Boundaries	5
Subtract.Block.Inlet	Index	4
	Interfaces	
	Part Surfaces	[Subtract.Block.Inlet]
	Type	Velocity Inlet
	Allow Per-Surface Values	false
	Tags	[]
Physics Conditions		
Flow Direction Specification	Method	Boundary-Normal
Reference Frame Specification	Option	Lab Frame
Turbulence Specification	Method	Intensity + Viscosity Ratio
Velocity Specification	Method	Magnitude + Direction
Physics Values		
Turbulence Intensity	Method	Constant

	Constant	Value	2.0
	Turbulent Viscosity Ratio	Method	Constant
	Constant	Value	10.0
	Velocity Magnitude	Method	Constant
	Constant	Value	42.0 m/s
Subtract.Block.outlet		Index	5
		Interfaces	
		Part Surfaces	[Subtract.Block.outlet]
		Type	Pressure Outlet
		Allow Per-Surface Values	false
		Tags	[]
	Physics Conditions		
	Backflow Specification	Direction	Boundary-Normal
		Pressure	Environmental
		Scalars	Specified
	Pressure Outlet Option	Option	None
	Reference Frame Specification	Option	Lab Frame
	Turbulence Specification	Method	Intensity + Viscosity Ratio
	Physics Values		
	Pressure	Method	Constant
	Constant	Value	1000.0 Pa
	Turbulence Intensity	Method	Constant
	Constant	Value	1.0
	Turbulent Viscosity Ratio	Method	Constant
	Constant	Value	10.0
Subtract.Block.Sides		Index	6
		Interfaces	
		Part Surfaces	[Subtract.Block.Sides]
		Type	Wall
		Allow Per-Surface Values	false
		Tags	[]
	Physics Conditions		
	Reference Frame Specification	Option	Region Reference Frame
	Shear Stress Specification	Method	No-Slip
	Tangential Velocity Specification	Method	Fixed
	Wall Surface Specification	Method	Smooth
	Physics Values		
	Blended Wall Function	E	9.0
		Kappa	0.42
Subtract.Block.Top&Bottom		Index	3
		Interfaces	
		Part Surfaces	[Subtract.Block.Top&Bottom]
		Type	Wall
		Allow Per-Surface Values	false
		Tags	[]
	Physics Conditions		
	Reference Frame Specification	Option	Region Reference Frame

Shear Stress Specification	Method	No-Slip
Tangential Velocity Specification	Method	Fixed
Wall Surface Specification	Method	Smooth
Physics Values		
Blended Wall Function	E	9.0
	Kappa	0.42
Subtract.TOYOTA GT.Faces	Index	2
	Interfaces	
	Part Surfaces	[Subtract.TOYOTA GT.Faces]
	Type	Wall
	Allow Per-Surface Values	false
	Tags	[]
Physics Conditions		
Reference Frame Specification	Option	Region Reference Frame
Shear Stress Specification	Method	No-Slip
Tangential Velocity Specification	Method	Fixed
Wall Surface Specification	Method	Smooth
Physics Values		
Blended Wall Function	E	9.0
	Kappa	0.42
Feature Curves	Part Curve Selection Priority	[Car:Feature Curve]
	Feature Curves	1
Feature Curve	Part Curves	[Subtract.TOYOTA GT.Edges, Subtract.Block.Block Curve]
	Tags	[]
Physics Conditions		
Initial Condition Option	Option	Use Continuum Values
Mass Source Option	Mass Source Option	false
Momentum Source Option	Momentum Source Option	None
Turbulence Source Option	Turbulence Source Option	None
Physics Values		
Axis	Coordinate System	Laboratory
	Origin	[0.0, 0.0, 0.0] m
	Direction	[0.0, 0.0, 1.0]
Motion Specification	Motion	Stationary
	Reference Frame	Lab Reference Frame
Representations		
Latest Surface/Volume	Representation	Volume Mesh
	Tags	[]
Geometry	Tags	[]
Automated Mesh.Remesh	Tags	[]
Latest Surface	Tags	[]
Volume Mesh	Cells	254293
	Interior Faces	1324236
	Vertices	950792
	Tags	[]
Finite Volume Regions		
Car	Cells	254293
	Interior Faces	1324236

	Vertices	950792
	Edges	0
Finite Volume		
Boundaries		
	Faces	623
Subtract.Block.Inlet		
	Faces	623
Subtract.Block.outlet		
	Faces	1616
Subtract.Block.Sides		
	Faces	1106
Subtract.Block.Top&Bottom		
Subtract.TOYOTA	Faces	21807
GT.Faces		
Cell Sets		
Contacts		
Parts		
Block	Metadata	{}
	Index	2
	Region	[]
	Contacts	[]
	Descriptions	[Root]
	Face Count	12
	Coordinate System	Laboratory
	Corner 1	[-3.0, -7.5, -3.5] m,m,m
	Corner 2	[3.0, 13.0, 5.0] m,m,m
	Tags	[]
Surfaces		
Inlet	Index	3
	Metadata	{}
	Boundary	[]
	Tags	[]
outlet	Index	4
	Metadata	{}
	Boundary	[]
	Tags	[]
Sides	Index	5
	Metadata	{}
	Boundary	[]
	Tags	[]
Top&Bottom	Index	2
	Metadata	{}
	Boundary	[]
	Tags	[]
Curves		
Block Curve	Index	2
	Feature Curve	[]
	Tags	[]
Subtract	Metadata	{}
	Index	3
	Region	[Car]
	Contacts	[]
	Descriptions	[Root, Automated Mesh.Remesh]
	Face Count	20822
	Operation	Subtract
	Tags	[]
TOYOTA GT	Metadata	{}
	Index	1
	Region	[]
	Contacts	[]
	Descriptions	[Root]

	Face Count	293994
	Tags	[]
Surfaces		
Faces	Index	1
	Metadata	{}
	Boundary	[]
	Tags	[]
Curves		
Edges	Index	1
	Feature Curve	[]
	Tags	[]
3D-CAD Models		
Operations		
Subtract	Link Output Part Name	false
	Input Parts	[TOYOTA GT, Block]
	Target Part	[Block]
	Perform CAD Boolean	true
	Output Parts	[Subtract]
	Tags	[]
Tessellation Options	Tessellation Options	Medium
Tolerance Type	Tolerance Type	Precise
	Tolerance	1.0E-5 m
Automated Mesh	Per-Part Meshing	false
	Mesher Execution Mode	Serial
	Input Parts	[Subtract]
	Perform Local Surface Meshing	false
	Preserve Surface Perimeters	None
	Verbose Output	false
	Tags	[]
Meshers		
Surface Remesher	Meshing Method	Triangle
	Perform Curvature Refinement	true
	Perform Proximity Refinement	true
	Perform Compatibility Refinement	false
	Create Aligned Meshes	true
	Minimum Face Quality	0.05
Polyhedral Mesher	Run Post Mesh Optimizer	true
Prism Layer Mesher	Stretching Function	Geometric Progression
	Distribution Mode	Stretch Factor
	Boundary March Angle	50.0
	Gap Fill Percentage	25.0
	Minimum Thickness Percentage	10.0
	Layer Reduction Percentage	50.0
	Concave Angle Limit	0.0
	Convex Angle Limit	360.0
	Near Core Layer Aspect Ratio	0.0
Default Controls		
Base Size	Base Size	0.5 m
	Base Size	0.5 m
CAD Projection	Project to CAD	true
Target Surface Size	Size Type	Relative to base
	Percentage of Base	100.0
	Absolute Size	0.5 m
Minimum Surface Size	Size Type	Relative to base
	Percentage of Base	10.0
	Absolute Size	0.05 m
Surface Curvature	Enable Curvature Deviation Distance	false
	# Pts/circle	36.0
	Max # Pts/circle	200.0

		Curvature Deviation Distance	0.01 m
	Surface Proximity	Search Floor	0.0 m
		# Points in gap	2.0
		Enable Search Ceiling	false
		Search Ceiling	1.0E10 m
	Surface Growth Rate	Surface Growth Rate	DEFAULT
		User Specified Value	1.3
	Number of Prism Layers	Number of Prism Layers	6
	Prism Layer Stretching	Prism Layer Stretching	1.0
	Prism Layer Total Thickness	Size Type	Relative to base
		Percentage of Base	25.0
		Absolute Size	0.125 m
	Volume Growth Rate	Volume Growth Rate	1.2
	Maximum Tet Size	Size Type	Relative to base
		Percentage of Base	10000.0
		Absolute Size	50.0 m
	Core Mesh Optimization	Optimization Cycles	1
		Quality Threshold	0.4
	Post Mesh Optimization	Optimize Boundary Vertices	false
		Optimize Cell Topology	false
	Custom Controls		
	Surface Control	Enable Control	true
		Controls Display Mode	All
		Part Surfaces	[Subtract.TOYOTA GT.Faces]
		Apply Only to Contacting Area	false
		Tags	[]
	Controls		
	Target Surface Size	Target Surface Size	Custom
	Minimum Surface Size	Minimum Surface Size	Custom
	Surface Curvature	Curvature	Custom
	Surface Proximity	Proximity	Parent
	Edge Proximity	Proximity	Parent
	Surface Growth Rate	Surface Growth Rate	Parent
	Surface Remeshing	Surface Remeshing	Parent
	Meshing Method	Meshing Method	Parent
	Prism Layers	Prism Layers	Parent
	Wake Refinement	Specify wake refinement options	false
	Values		
	Target Surface Size	Size Type	Relative to base
		Percentage of Base	6.0
		Absolute Size	0.03 m
	Minimum Surface Size	Size Type	Relative to base
		Percentage of Base	5.0
		Absolute Size	0.025 m
	Surface Curvature	Enable Curvature Deviation Distance	true
		# Pts/circle	90.0
		Max # Pts/circle	200.0
		Curvature Deviation Distance	0.01 m
Descriptions		Number of Children	4

Root	Described Parts	[TOYOTA GT, Block, Subtract]
Automated Mesh.Remesh	Described Parts	[Subtract]
	Faces	48884
	Vertices	24446
Latest Surface	Described Parts	[TOYOTA GT, Block, Subtract]
	Faces	342890
	Vertices	171453
	Preview Mesh Operation Parts	false
Latest Surface/Volume	Described Parts	[TOYOTA GT, Block, Subtract]
Coordinate Systems		
Laboratory	Tags	[]
Local Coordinate Systems		
Tables	Tables	0
Tags		
Units	Preferred System	Systeme Internationale
Custom Trees		
Filters		
Field Functions		
Simulation Operations	Selected	[]
Parameters		
Volume Shapes		
Data Set Functions	Data Directory	function_data
Update Events	Event Count	0
	Event Names	
Time Scales		
User Code		
Data Focus		
Layouts		
default		
Data Mappers	Use Legacy Volume Mapper	false
Motions		
Stationary	Tags	[]
Reference Frames		
Lab Reference Frame	Tags	[]
Screenplays		
Derived Parts	Derived Parts	2
Y section	Origin	[0.0, 2.75, 0.75] m,m,m
	Coordinate System	Laboratory
	Normal	[1.0, 0.0, 0.0] m,m,m
	Parts	[Car]
	Section Mode	SINGLE
	Displayed Index	-1
	Tags	[]
Single section	Offset	0.0 m
Z section	Origin	[0.0, 2.75, 0.5] m,m,m
	Coordinate System	Laboratory
	Normal	[0.0, 0.0, 1.0] m,m,m
	Parts	[Car]
	Section Mode	SINGLE
	Displayed Index	-1
	Tags	[]
Single section	Offset	0.0 m
Summaries		
Monitors	Monitors	9
	Monitors To Print	[Continuity, X-momentum, Y-momentum, Z-momentum, Energy, Tke, Tdr]
	Output Direction	Horizontal
	Heading Print Frequency	10

Iteration	Maximum Plot Samples	5000
	Tags	[]
Physical Time	Maximum Plot Samples	5000
	Tags	[]
Reports	Reports	2
Drag	Units	N
	Coordinate System	Laboratory
	Direction	[0.0, -1.0, 0.0]
	Force Option	Pressure + Shear
	Reference Pressure	0.0 Pa
	Number of Bands	0
	Parts	[Car: Subtract.TOYOTA GT.Faces]
	Representation	Volume Mesh
	Smooth Values	false
	Tags	[]
Lift	Units	N
	Coordinate System	Laboratory
	Direction	[0.0, 1.0, 1.0]
	Force Option	Pressure + Shear
	Reference Pressure	0.0 Pa
	Number of Bands	0
	Parts	[Car: Subtract.TOYOTA GT.Faces]
	Representation	Volume Mesh
	Smooth Values	false
	Tags	[]
Solvers		
Implicit Unsteady	Time-Step	0.001 s
	Temporal Discretization	1st-order
	Freeze Time	false
Stopping Criteria	Verbose	false
Partitioning	Solver Frozen	false
	Partitioning Method	Per-Continuum
Wall Distance	Solver Frozen	false
	Verbosity	0
	Minimum Tree Size Threshold	500000
Segregated Flow	Pressure Gradient Inverse Distance Weighting	true
	Freeze Flow	false
	Reconstruction Frozen	false
	Reconstruction Zeroed	false
	Enable Enhanced Stability Treatment	false
	Temporary Storage Retained	false
	Pressure Corrections: Bad Cell Minimum Scaling	0.8
	Pressure Corrections: Acceptable Cell Volume Change	0.001
	Velocity Corrections: Maximum Unlimited Velocity	20.0 m/s
	Velocity Corrections: Acceptable Velocity Increase Rate [<1]	0.15
	Continuity Initialization	false
Velocity	Under-Relaxation Factor	0.8
	Dynamic Local Under-Relaxation	false
Under-Relaxation Factor Ramp	Ramp Method	No Ramp
AMG Linear Solver	Max Cycles	30
	Verbosity	NONE
	Enable Direct Solver	false
	Maximum Direct Solver Equations	32
	Convergence Tolerance	0.1

		Epsilon	0.0
		Cycle Type	Flex Cycle
		Group Size Control	Auto
		Group Size	4
		Relaxation Scheme	Gauss-Seidel
		Acceleration Method	None
		Scaling	Disabled
	Flex Cycle	Restriction Tolerance	0.9
		Prolongation Tolerance	0.5
		Sweeps	1
	Pressure	Number of Non-Orthogonality Correctors	0
		Under-Relaxation Factor	0.2
		Pressure Reference Location	Automatic Selection
	Under-Relaxation Factor Ramp	Ramp Method	No Ramp
	AMG Linear Solver	Max Cycles	30
		Verbosity	NONE
		Enable Direct Solver	false
		Maximum Direct Solver Equations	32
		Convergence Tolerance	0.1
		Epsilon	0.0
		Cycle Type	V Cycle
		Group Size Control	Auto
		Group Size	4
		Relaxation Scheme	Gauss-Seidel
		Acceleration Method	Bi Conjugate Gradient Stabilized
		Scaling	Auto
	V Cycle	Pre-Sweeps	1
		Post-Sweeps	1
		Max Levels	50
	Segregated Energy	Solver Frozen	false
		Reconstruction Frozen	false
		Reconstruction Zeroed	false
		Temporary Storage Retained	false
		Fluid Under-Relaxation Factor	0.9
		Solid Under-Relaxation Factor	0.99
		Enable High-Accuracy Temporal Discretization	false
	Fluid Under-Relaxation Factor Ramp	Ramp Method	No Ramp
	Solid Under-Relaxation Factor Ramp	Ramp Method	No Ramp
	AMG Linear Solver	Max Cycles	30
		Verbosity	NONE
		Enable Direct Solver	false
		Maximum Direct Solver Equations	32
		Convergence Tolerance	0.1
		Epsilon	0.0
		Cycle Type	V Cycle
		Group Size Control	Auto
		Group Size	4
		Relaxation Scheme	Gauss-Seidel
		Acceleration Method	None
		Scaling	Disabled
	V Cycle	Pre-Sweeps	1
		Post-Sweeps	1
		Max Levels	50
	K-Epsilon Turbulence	Solver Frozen	false
		Reconstruction Frozen	false

		Reconstruction Zeroed	false
		Temporary Storage Retained	false
		Under-Relaxation Factor	0.8
		Boundary Layer Initialization	false
	Under-Relaxation Factor	Ramp Method	No Ramp
Ramp			
	AMG Linear Solver	Max Cycles	30
		Verbosity	NONE
		Enable Direct Solver	false
		Maximum Direct Solver Equations	32
		Convergence Tolerance	0.1
		Epsilon	0.0
		Cycle Type	Flex Cycle
		Group Size Control	Auto
		Group Size	4
		Relaxation Scheme	Gauss-Seidel
		Acceleration Method	None
		Scaling	Disabled
	Flex Cycle	Restriction Tolerance	0.9
		Prolongation Tolerance	0.5
		Sweeps	1
	K-Epsilon Turbulent	Solver Frozen	false
Viscosity			
		Under-Relaxation Factor	1.0
		Maximum Ratio	100000.0
	Stopping Criteria	Verbose	false
	Maximum Inner Iterations	Enabled	true
		Maximum Inner Iterations	3
		Logical Rule	Or
		Criterion Satisfied	false
		Tags	[]
	Maximum Physical Time	Enabled	true
		Maximum Physical Time	1.0 s
		Logical Rule	Or
		Criterion Satisfied	false
		Tags	[]
	Maximum Steps	Enabled	true
		Maximum Steps	200
		Logical Rule	Or
		Criterion Satisfied	false
		Tags	[]
	Stop File	Enabled	true
		Stop Inner Iterations	true
		Path	ABORT
		Logical Rule	Or
		Criterion Satisfied	false
		Tags	[]
	Solution Histories		
	Solution Views		
	Current Solution	Iteration	37
		Time Step	12
		Solution Time	0.013000000000000005
		Tags	[]
	Layout Views		

Solution

Accumulated CPU Time over all processes (s)	114.11500000000008
Elapsed Time (s)	57.061009000011836
Time Level	12

Solution Time	0.013000000000000005